Python Lists Assessment Test

Comprehensive Multiple-Choice Test & Answer Key

30 Questions • 1 Mark Each

Instructions: Select the correct answer for each question. The complete answer key with detailed explanations is provided at the end of the test document.

1. Which symbol is used to create a list in Python?

A. {}
B. ()
C. []
D. <>

2. What is the output of `type([10, 20, 30])`?

A. <class 'tuple'>
B. <class 'list'>
C. <class 'dict'>
D. <class 'set'>

3. Which statement about Python lists is correct?

A. Lists are immutable
B. Lists cannot contain duplicates
C. Lists maintain insertion order
D. Lists can store only integers

4. What is the output of `x = [10, 20, 30]; print(x[1])`?

A. 10
B. 20
C. 30
D. 1

5. What is the output of `x = [10, 20, 30, 40]; print(x[-1])`?

A. 10
B. 20
C. 30
D. 40

6. What is the output of `x = [1, 2, 3]; x[1] = 9; print(x)`?

A. [9, 2, 3]
B. [1, 9, 3]
C. [1, 2, 9]
D. Error

7. What is the output of `x = [1, 2]; x.append([3, 4]); print(len(x))`?

A. 2
B. 3
C. 4
D. Error

16. What is the output of `x = [1, 2, 3]; print(x[::-1])`?

A. [1, 2, 3]
B. [3, 2, 1]
C. [2, 1, 3]
D. []

17. What is the output of `print([1, 2] + [3, 4])`?

A. [4, 6]
B. [1, 2, 3, 4]
C. [[1, 2], [3, 4]]
D. Error

18. What is the output of `print([1, 2] * 2)`?

A. [2, 4]
B. [1, 2, 2]
C. [1, 2, 1, 2]
D. [1, 1, 2, 2]

19. What is the output of `print(3 in [1, 2, 3])`?

A. True
B. False
C. 3
D. Error

20. What is the output of `x = [4, 1, 3]; y = sorted(x); print(x)`?

A. [1, 3, 4]
B. [4, 1, 3]
C. None
D. Error

21. What is the output of `x = [3, 1, 2]; print(x.sort())`?

A. [1, 2, 3]
B. [3, 1, 2]
C. None
D. True

22. What is the output of `a = [1, 2]; b = a; b[0] = 9; print(a)`?

A. [1, 2]
B. [9, 2]
C. [1, 9]
D. Error

8. What is the output of `x = [1, 2]; x.extend([3, 4]); print(x)` ?

A. [1, 2, [3, 4]]
B. [1, 2, 3, 4]
C. [[1, 2], [3, 4]]
D. [3, 4]

9. What is the output of `x = [10, 20, 30]; x.insert(1, 15); print(x)` ?

A. [15, 10, 20, 30]
B. [10, 15, 20, 30]
C. [10, 20, 15, 30]
D. [10, 20, 30, 15]

10. What does `remove(20)` do when 20 appears twice in a list?

A. Removes both occurrences
B. Removes the first occurrence
C. Removes the last occurrence
D. Removes the element at index 20

11. What is the output of `x = [5, 10, 15]; print(x.pop())` ?

A. 5
B. 10
C. 15
D. None

12. What is the output of `x = [1, 2, 3]; x.clear(); print(x)` ?

A. None
B. [1, 2, 3]
C. []
D. Error

13. What is the output of `x = [2, 4, 2, 6]; print(x.count(2))` ?

A. 1
B. 2
C. 3
D. 4

14. What is the output of `x = [10, 20, 30, 20]; print(x.index(20))` ?

A. 1
B. 2
C. 3
D. 20

15. What is the output of `x = [10, 20, 30, 40]; print(x[1:3])` ?

A. [10, 20, 30]
B. [20, 30]
C. [20, 30, 40]
D. [10, 20]

23. What is the output of `a = [1, 2]; b = a.copy(); b[0] = 9; print(a)` ?

A. [9, 2]
B. [1, 2]
C. [1, 9]
D. [9, 9]

24. Given `x = [[10, 20], [30, 40]]`, what is `x[1][0]` ?

A. 10
B. 20
C. 30
D. 40

25. What is the output of `print([n*n for n in range(1, 4)])` ?

A. [1, 2, 3]
B. [1, 4, 9]
C. [1, 4, 9, 16]
D. [0, 1, 4]

26. What is the output of `print([n for n in range(6) if n % 2 == 0])` ?

A. [1, 3, 5]
B. [0, 2, 4]
C. [2, 4, 6]
D. [0, 1, 2, 3, 4, 5]

27. What does `len([10, [20, 30], 40])` return?

A. 2
B. 3
C. 4
D. 5

28. What is printed by `for i, v in enumerate(['a', 'b']): print(i, v)` ?

A. 1 a then 2 b
B. 0 a then 1 b
C. a 0 then b 1
D. 0 b then 1 a

29. What does `zip([1, 2, 3], ['a', 'b'])` pair before it stops?

A. (1, 'a'), (2, 'b'), (3, None)
B. (1, 'a'), (2, 'b')
C. (1, 'a'), (2, 'b'), (3, '')
D. It raises an error because lengths differ

30. What is the output of `x = [1, 2, 3]; result = x.append(4); print(result)` ?

A. [1, 2, 3, 4]
B. 4
C. None
D. True

Answer Key & Detailed Explanations

Q1: C. []
Square brackets `[]` define a list literal in Python. `{}` creates a dict/set, `()` creates a tuple.

Q2: B. <class 'list'>
The `type()` function returns the class/type of the object, which is `list`.

Q3: C. Lists maintain insertion order
Python lists are mutable, ordered (preserve insertion order), and allow duplicate elements of mixed data types.

Q4: B. 20
Python uses 0-based indexing. Index 0 is 10, index 1 is 20.

Q5: D. 40
Index `-1` accesses the last element of the list.

Q6: B. [1, 9, 3]
Lists are mutable; assigning to `x[1]` replaces element 2 at index 1 with 9.

Q7: B. 3
`append()` adds the list `[3, 4]` as a single nested element, giving `[1, 2, [3, 4]]` (length 3).

Q8: B. [1, 2, 3, 4]
`extend()` iterates over its argument, appending each element individually.

Q9: B. [10, 15, 20, 30]
`insert(1, 15)` inserts 15 at index 1, shifting remaining items to the right.

Q10: B. Removes the first occurrence
`list.remove(x)` removes only the first matching item found from left to right.

Q11: C. 15
`pop()` removes and returns the last element (15) by default.

Q12: C. []
`clear()` removes all elements in-place, leaving an empty list `[]`.

Q13: B. 2
`count(2)` returns the number of times 2 appears in the list (2 times).

Q16: B. [3, 2, 1]
`[::-1]` slices the list with a step of -1, reversing its elements.

Q17: B. [1, 2, 3, 4]
The + operator concatenates two lists into a new single list.

Q18: C. [1, 2, 1, 2]
The * operator repeats the list items the specified number of times.

Q19: A. True
The `in` membership operator evaluates whether an element exists in the list.

Q20: B. [4, 1, 3]
`sorted()` returns a new sorted list, leaving the original list x unmodified.

Q21: C. None
`list.sort()` sorts in-place and returns `None`.

Q22: B. [9, 2]
`b = a` creates a reference to the same list object; mutating b affects a.

Q23: B. [1, 2]
`copy()` creates a shallow copy. Modifying b does not affect original list a.

Q24: C. 30
`x[1]` gives `[30, 40]`; accessing index 0 yields `30`.

Q25: B. [1, 4, 9]
`range(1, 4)` produces 1, 2, 3. Squares are 1*1=1, 2*2=4, 3*3=9.

Q26: B. [0, 2, 4]
`range(6)` produces 0 to 5. Even numbers (`n % 2 == 0`) are 0, 2, 4.

Q27: B. 3
The outer list contains 3 items: integer 10, nested list `[20, 30]`, and integer 40.

Q28: B. 0 a then 1 b
`enumerate()` yields (index, value) tuples starting at index 0.

Q14: A. 1

`index(20)` returns the 0-based index of the first occurrence of 20, which is at index 1.

Q15: B. [20, 30]

Slicing `[1:3]` starts at index 1 and goes up to (excluding) index 3.

Q29: B. (1, 'a'), (2, 'b')

`zip()` stops as soon as the shortest input iterable is exhausted.

Q30: C. None

`append()` modifies the list in-place and returns None.